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$$\begin{aligned} & \lim_{x \rightarrow e} \frac{\sqrt{x} - \sqrt{e}}{\ln x - 1} \\ & \stackrel{H}{=} \lim_{x \rightarrow e} \frac{\frac{d}{dx}(\sqrt{x} - \sqrt{e})}{\frac{d}{dx}(\ln x - 1)} \\ & = \lim_{x \rightarrow e} \frac{\left(\frac{1}{2\sqrt{x}}\right)}{\left(\frac{1}{x}\right)} \\ & = \lim_{x \rightarrow e} \frac{x}{2\sqrt{x}} \\ & = \lim_{x \rightarrow e} \frac{\sqrt{x} \cdot \sqrt{x}}{2\sqrt{x}} \\ & = \lim_{x \rightarrow e} \frac{\sqrt{x}}{2} \\ & = \frac{\sqrt{e}}{2} \end{aligned}$$

References